Lecture 11

Kinematics of Rigid Bodies III

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MC2103

Fall 2026

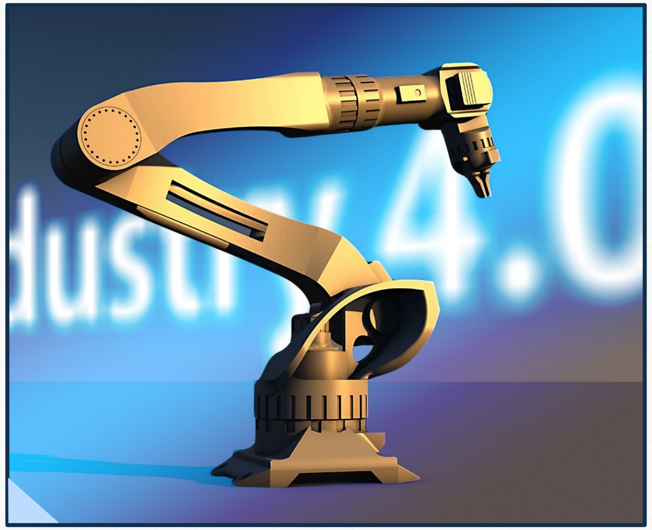
Lecture 11-1

https://youtu.be/gaQrJlxJK3M

Lecture 11-2

Introduction

In Week 10, we studied direction cosine matrix and rotation matrix.
In this week, we will study vector differentiation formula, angular velocity, angular acceleration, and linear velocity and acceleration.



Differentiating Vectors in Different Frames

Vector differentiation

- We learned that differentiating one vector in one reference frame may not be the same as differentiating the vector in different reference frame.
- Luckily, the two results are related.
- Here, we present a formula for this matter without providing the proof.

Differentiating Vectors in Different Frames

Vector differentiation

- Given two reference frames A and B, any vector $\vec{v}$ can be differentiated with respect to time in the following manner.

$$\frac{{}^{A}d\vec{v}}{dt} = \frac{{}^{B}d\vec{v}}{dt} + {}^{A}\vec{\omega}^{B} \times \vec{v}$$

where ${}^{A}\vec{\omega}^{B}$ is the angular velocity of reference frame B with respect to reference frame A.

Simple Angular Velocity

Obtaining an angular velocity of a rigid body in 3D space

- Unlike rigid bodies in 2D space, obtaining an angular velocity of a rigid body in 3D space is not straightforward and intuitive.

FYI, given two reference frames A and B along with coordinate systems $\hat{a}_1\hat{a}_2\hat{a}_3$, and $\hat{b}_1\hat{b}_2\hat{b}_3$,

$$ {}^{A}\vec{\omega}^{B} = \left(\frac{{}^{A}\hat{b}_2}{dt}\cdot\hat{b}_3\right)\hat{b}_1 + \left(\frac{{}^{A}\hat{b}_3}{dt}\cdot\hat{b}_1\right)\hat{b}_2 + \left(\frac{{}^{A}\hat{b}_1}{dt}\cdot\hat{b}_2\right)\hat{b}_3 $$

Simple Angular Velocity

Obtaining an angular velocity of a rigid body in 3D space

- Instead, an angular velocity of a reference frame B rotating about a single axis $\hat{k}$ fixed in a reference frame A can be expressed as

$$^{A}\vec{\omega}^{B} = \dot{\theta}\hat{k}$$

where $\dot{\theta}$ is the rate change of angular displacement θ.

Addition of Angular Velocities

Addition theorem of angular velocities

- General rotational movements involve rotations about multiple axes.
- Given multiple reference frames $A, A_1, A_2, \cdots, A_{n-1}, A_n$, and B, we have the following formula (without proof)

$$ {}^{A}\vec{\omega}^{B} = {}^{A}\vec{\omega}^{A_1} + {}^{A_1}\vec{\omega}^{A_2} + \cdots + {}^{A_{n-1}}\vec{\omega}^{A_n} + {}^{A_n}\vec{\omega}^{B} $$

- This relation is called addition theorem of angular velocities.

Addition of Angular Velocities

Addition theorem of angular velocities

- Using the addition theorem and simple angular velocities, complicated angular velocities can be built up.
- Please note that addition theorem does not work for angular acceleration since the addition theorem is a byproduct of vector differentiation formula which contains an angular velocity.
    - FYI, angular acceleration can be computed by differentiating the angular velocity.

Lecture 11-3

Two Points Fixed on a Reference Frame

Linear velocity and acceleration

- Since we have studied angular velocities and accelerations, we now study linear velocity and acceleration.
- First of all, consider two fixed points P and Q on a reference frame A.

Two Points Fixed on a Reference Frame

Linear velocity and acceleration

- If we know the linear velocity and acceleration of point Q, the linear velocity and acceleration of point P can be computed as follows.

$$\vec{v}^P = \vec{v}^Q + \vec{\omega}^A \times \overrightarrow{QP}$$
$$\vec{a}^P = \vec{a}^Q + \vec{\alpha}^A \times \overrightarrow{QP} + \vec{\omega}^A \times \left(\vec{\omega}^A \times \overrightarrow{QP}\right)$$

where $\vec{\omega}^A$ and $\vec{\alpha}^A$ are the angular velocity and angular acceleration of reference frame A. Here the base reference frame is omitted since it is obvious in the context. Usually, it is a Newtonian reference frame N.

One Point Moving on a Reference Frame

- **Consider a moving point P and on a reference frame A.**
  - Also, consider a fixed point P^* on the reference frame A that coincides with the moving point P.

One Point Moving on a Reference Frame

Consider a moving point P and on a reference frame A.

- If we know the linear velocity and acceleration of the fixed point P^*, the linear velocity and acceleration of the moving point P can be computed as follows.

$$\vec{v}^P = \vec{v}^{P^*} + {}^A\vec{v}^P$$
$$\vec{a}^P = \vec{a}^{P^*} + {}^A\vec{a}^P + 2\vec{\omega}^A \times {}^A\vec{v}^P$$

where $\vec{\omega}^A$ is the angular velocity of reference frame A. ${}^A\vec{v}^P$ is the relative velocity of the point P observed from the reference frame A. Here the base reference frame is omitted since it is obvious in the context. Usually, it is a Newtonian reference frame N.

Summary of Formula

Formula

- Vector differentiation formula

$$\frac{{}^{A}d\vec{v}}{dt} = \frac{{}^{B}d\vec{v}}{dt} + {}^{A}\vec{\omega}^{B} \times \vec{v}$$

- Simple angular velocity

$${}^{A}\vec{\omega}^{B} = \dot{\theta}\hat{k}$$

- Addition theorem of angular velocities

$${}^{A}\vec{\omega}^{B} = {}^{A}\vec{\omega}^{A_1} + {}^{A_1}\vec{\omega}^{A_2} + \cdots + {}^{A_{n-1}}\vec{\omega}^{A_n} + {}^{A_n}\vec{\omega}^{B}$$

Summary of Formula

Formula

- Two points fixed on one reference frame

$$\vec{v}^P = \vec{v}^Q + \vec{\omega}^A \times \overrightarrow{QP}$$
$$\vec{a}^P = \vec{a}^Q + \vec{\alpha}^A \times \overrightarrow{QP} + \vec{\omega}^A \times \left(\vec{\omega}^A \times \overrightarrow{QP}\right)$$

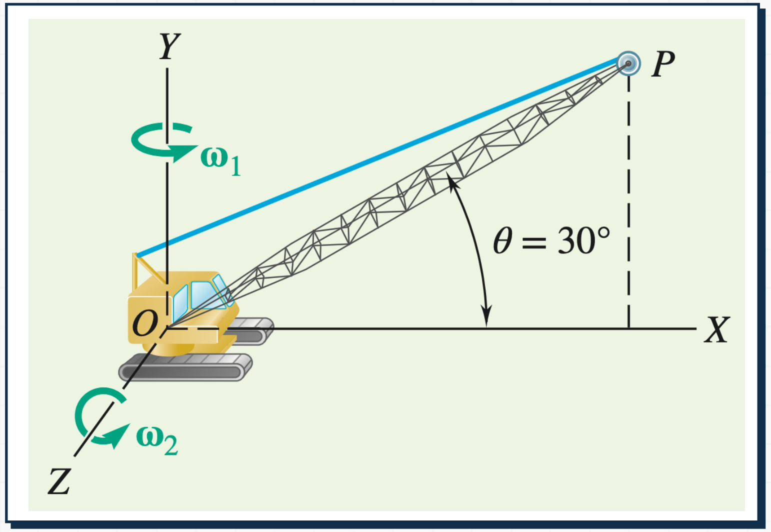
- One point moving on one reference frame

$$\vec{v}^P = \vec{v}^{P^*} + {}^A\vec{v}^P$$
$$\vec{a}^P = \vec{a}^{P^*} + {}^A\vec{a}^P + 2\vec{\omega}^A \times {}^A\vec{v}^P$$

Lecture 11-4

Sample Problem 1

Consider the following crane.



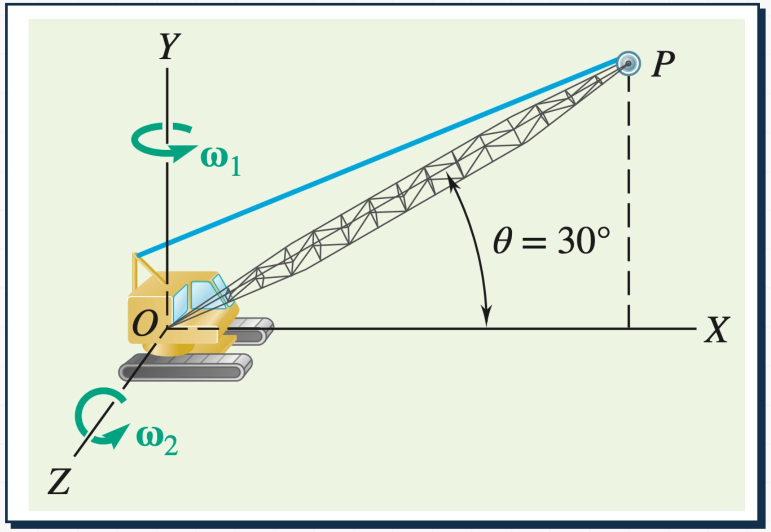
Crane rotates horizontally with a constant angular velocity $\omega_1 = 0.3$ rad/s.

Simultaneously, the boom is being raised with a constant angular velocity $\omega_2 = 0.5$ rad/s relative to the cab.

$\overline{OP} = l = 12\text{m}$

Sample Problem 1

Consider the following crane.



- Determine
  - The angular velocity of the boom
  - The angular acceleration of the boom
  - The velocity of the tip of the boom
  - The acceleration of the tip of the boom

Sample Problem 1

Consider the following crane.

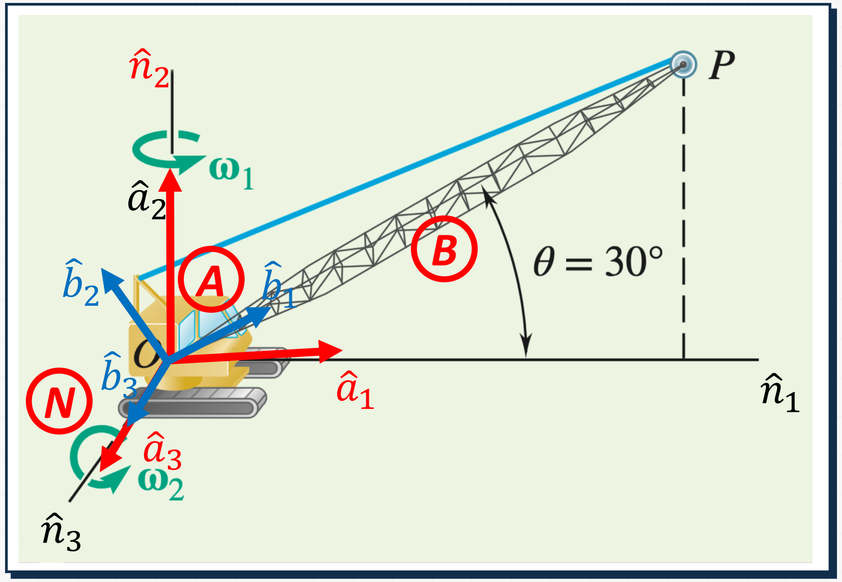
Strategy

- Identify appropriate reference frames. It is useful to have only one degree of freedom between each reference frame to use the simple angular velocity.
- Assign appropriate coordinate systems to each reference frame.
- Build the direction cosine table.
- Obtain the relative angular velocities between reference frames.
- Compute the angular accelerations by differentiating the angular velocities of interest using vector differentiation formula.
- Use two-point relation to compute linear velocity and acceleration of the tip.

GIST

Sample Problem 1

Define reference frames and coordinate systems.



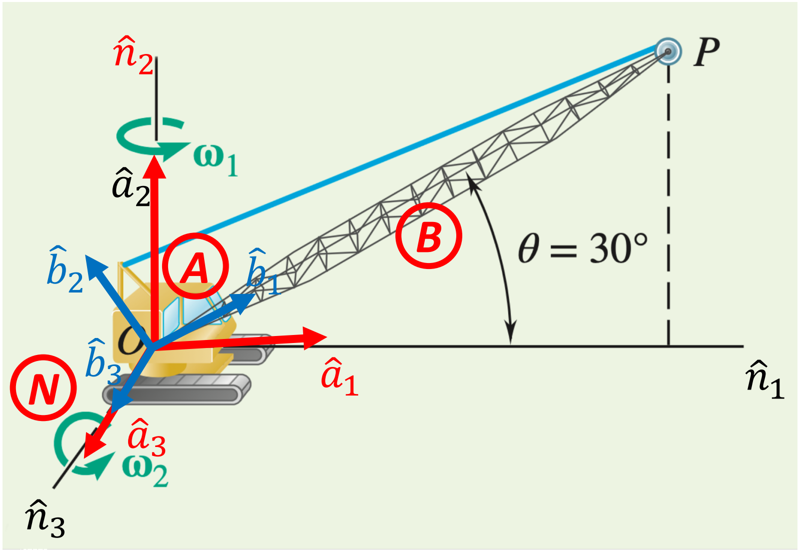
Inertial reference frame: N
with $\hat{n}_1\hat{n}_2\hat{n}_3$

Reference frame of cab: A
with $\hat{a}_1\hat{a}_2\hat{a}_3$

Reference frame of boom: B
with $\hat{b}_1\hat{b}_2\hat{b}_3$

Sample Problem 1

Build direction cosine tables.



Cab rotates with angle θ about y axis with respect to the ground.

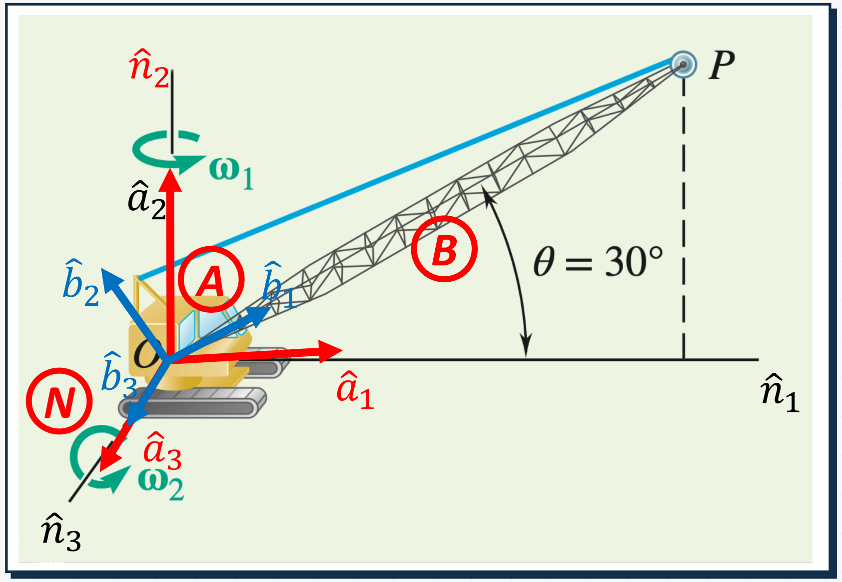
Boom rotates with angle φ about z axis with respect to the cab.

	$\hat{a}_1$	$\hat{a}_2$	$\hat{a}_3$
$\hat{n}_1$	$\cos\theta$	0	$\sin\theta$
$\hat{n}_2$	0	1	0
$\hat{n}_3$	$-\sin\theta$	0	$\cos\theta$

	$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$
$\hat{a}_1$	$\cos\varphi$	$-\sin\varphi$	0
$\hat{a}_2$	$\sin\varphi$	$\cos\varphi$	0
$\hat{a}_3$	0	0	1

Sample Problem 1

Obtain relative angular velocities.



Cab rotates with angle θ about y axis with respect to the ground.

$$^{N}\vec{\omega}^{A} = \omega_1 \hat{n}_2 = \omega_1 \hat{a}_2$$

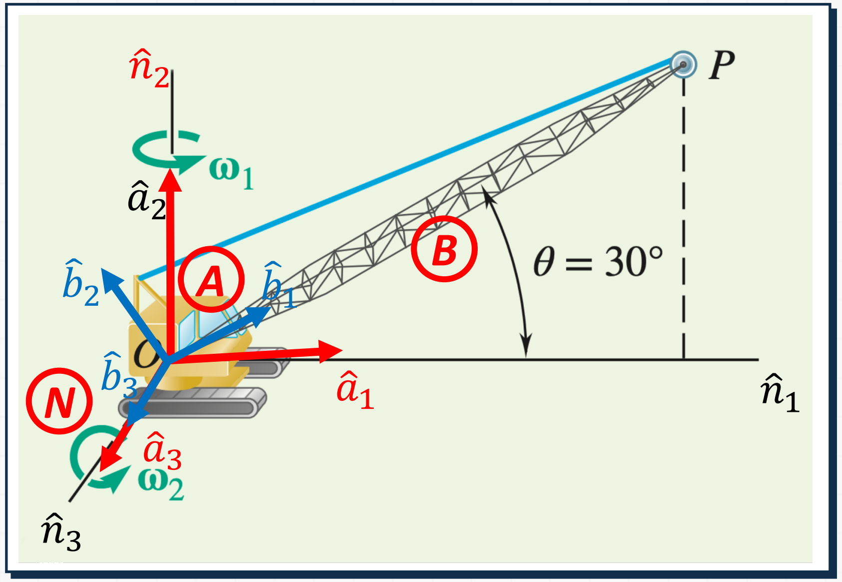
Boom rotates with angle φ about z axis with respect to the cab.

$$^{A}\vec{\omega}^{B} = \omega_2 \hat{a}_3 = \omega_2 \hat{b}_3$$

GIST

Sample Problem 1

Obtain relative angular velocities.

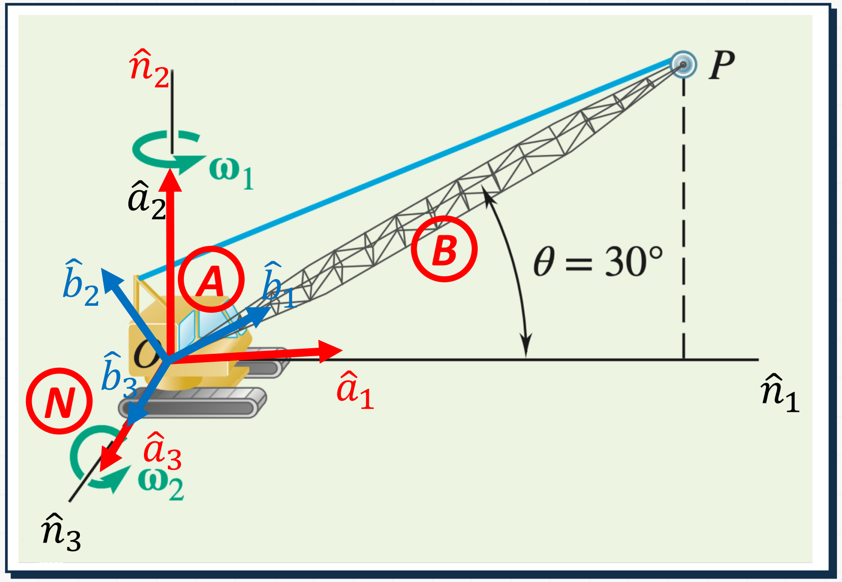


Angular velocity of Boom with respect to the ground.

$$^{N}\vec{\omega}^{B} = {}^{N}\vec{\omega}^{A} + {}^{A}\vec{\omega}^{B}$$
$$= \omega_1 \hat{a}_2 + \omega_2 \hat{a}_3$$

Sample Problem 1

Obtain relative angular velocities.



Cab rotates with angle θ about y axis with respect to the ground.

$$^{N}\vec{\omega}^{A} = \omega_1 \hat{n}_2 = \omega_1 \hat{a}_2$$

Boom rotates with angle φ about z axis with respect to the cab.

$$^{A}\vec{\omega}^{B} = \omega_2 \hat{a}_3 = \omega_2 \hat{b}_3$$

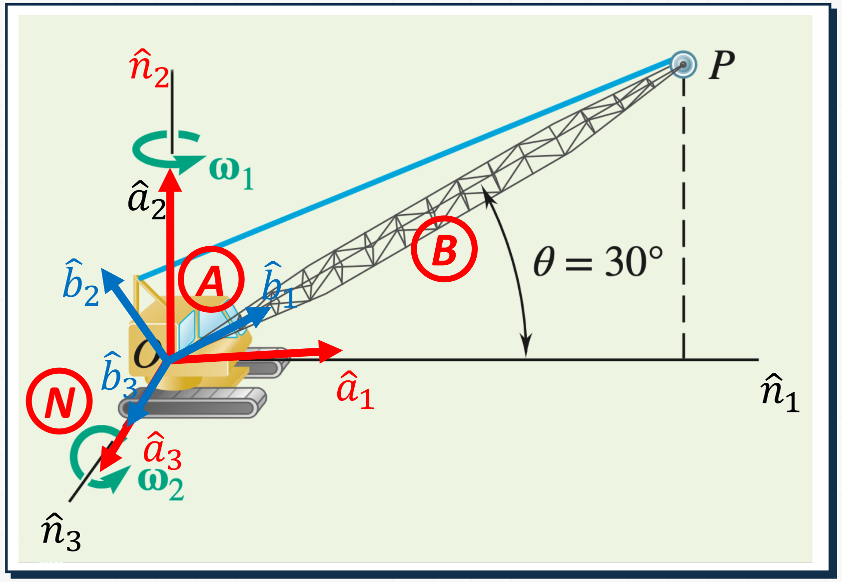
Angular velocity of Boom with respect to the ground.

$$^{N}\vec{\omega}^{B} = {}^{N}\vec{\omega}^{A} + {}^{A}\vec{\omega}^{B} = \omega_1 \hat{a}_2 + \omega_2 \hat{a}_3$$

GIST

Sample Problem 1

Compute angular accelerations.



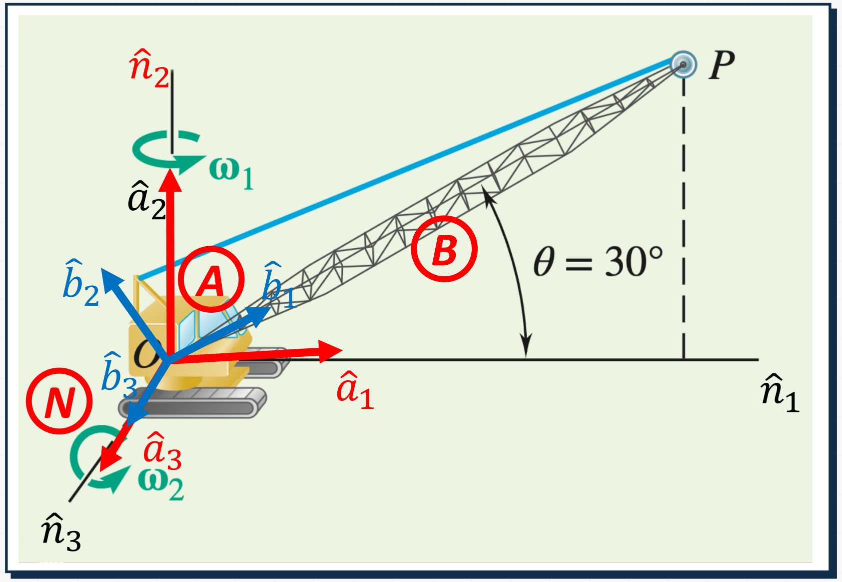
Compute the angular acceleration of the Boom using vector differentiation formula.

$$
\begin{aligned}
{}^{N}\vec{\alpha}^{B} &= \frac{{}^{N}d}{dt}\left({}^{N}\vec{\omega}^{B}\right) \\
&= \frac{{}^{A}d}{dt}\left({}^{N}\vec{\omega}^{B}\right) + {}^{N}\vec{\omega}^{A} \times {}^{N}\vec{\omega}^{B} \\
&= \frac{{}^{A}d}{dt}\left({}^{N}\vec{\omega}^{B}\right) + {}^{N}\vec{\omega}^{A} \times {}^{A}\vec{\omega}^{B} \\
&= \dot{\omega}_1\hat{a}_2 + \dot{\omega}_2\hat{a}_3 + \omega_1\hat{a}_2 \times \omega_2\hat{a}_3 \\
&= 0 + \omega_1\omega_2\hat{a}_1 \\
&= \omega_1\omega_2\hat{a}_1 \\
&= 0.15\hat{a}_1
\end{aligned}
$$

GIST

Sample Problem 1

Compute the velocity of the tip.



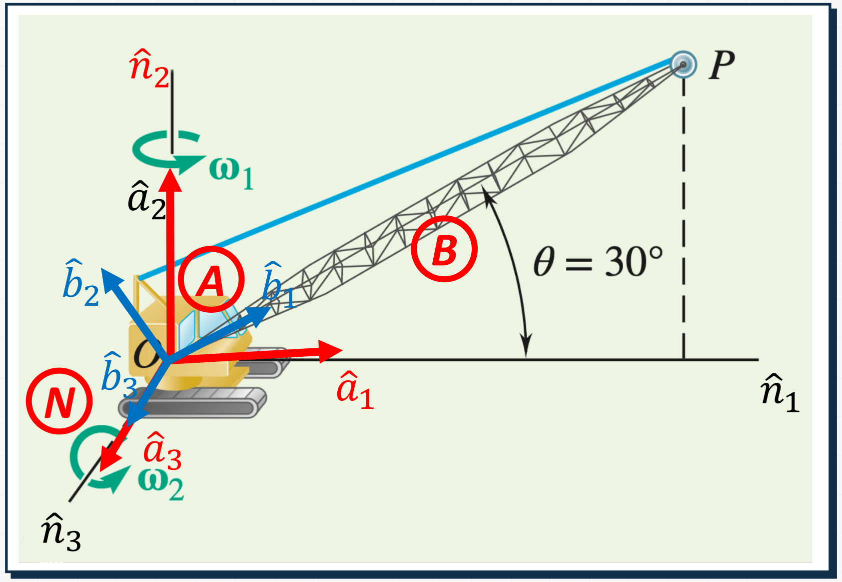
We know the velocity of the point O, which is zero.
Therefore, it is better to use two-point relation.

$$\begin{aligned}
{}^N\vec{v}^P &= {}^N\vec{v}^O + {}^N\vec{\omega}^B \times \overrightarrow{OP} \\
&= 0 + (\omega_1\hat{a}_2 + \omega_2\hat{a}_3)\times l\hat{b}_1 \\
&= (\omega_1\hat{a}_2 + \omega_2\hat{b}_3)\times l\hat{b}_1 \\
&= \omega_1 l\hat{a}_2\times\hat{b}_1 + \omega_2 l\hat{b}_3\times\hat{b}_1 \\
&= \omega_1 l\hat{a}_2\times\hat{b}_1 + \omega_2 l\hat{b}_2 \\
&= \omega_1 l(\sin\varphi\,\hat{b}_1 + \cos\varphi\,\hat{b}_2)\times\hat{b}_1 \\
&= +\omega_2 l\hat{b}_2 \\
&= -\omega_1 l\cos\varphi\,\hat{b}_3 + \omega_2 l\hat{b}_2 \\
&= 6\hat{b}_2 - 3.12\hat{b}_3
\end{aligned}$$

GIST

Sample Problem 1

Compute the acceleration of the tip.



We know the acceleration of the point O, which is zero. Therefore, it is better to use two-point relation.

$$
\begin{aligned}
{}^N\vec{a}^P &= {}^N\vec{a}^O + {}^N\vec{\alpha}^B \times \overrightarrow{OP} + {}^N\vec{\omega}^B \times \left({}^N\vec{\omega}^B \times \overrightarrow{OP}\right) \\
&= 0 + \omega_1\omega_2\hat{a}_1 \times l\hat{b}_1 + (\omega_1\hat{a}_2 + \omega_2\hat{a}_3) \\
&= \times\left((\omega_1\hat{a}_2 + \omega_2\hat{a}_3) \times l\hat{b}_1\right) \\
&= \cdots \\
&= -3.81\hat{b}_1 + 0.47\hat{b}_2 + 1.8\hat{b}_3 \\
&= or \\
&= -3.53\hat{n}_1 - 1.5\hat{n}_2 + 1.8\hat{n}_3
\end{aligned}
$$

Sample Problem 1

Reflect and Think

The base of the cab acts as the fixed point of the motion.

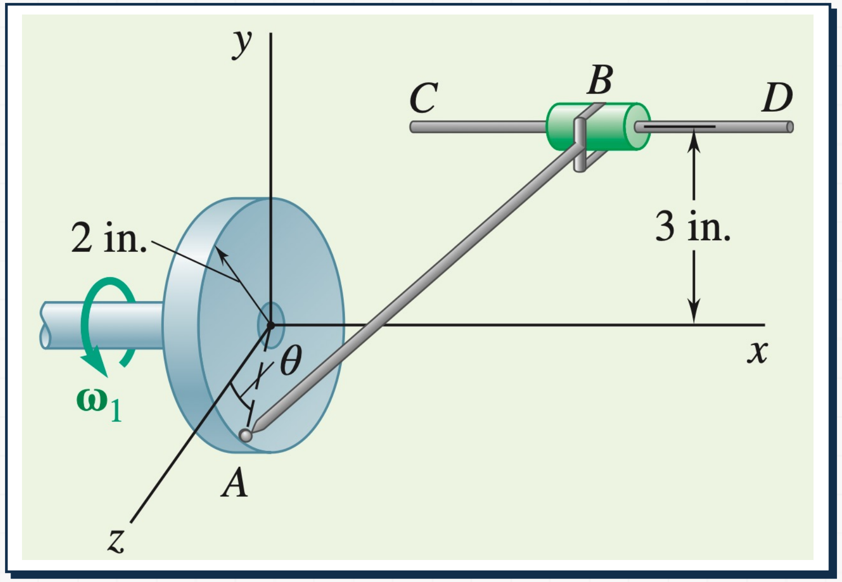
Even though both components of angular velocity are constant, there is an angular acceleration due to the change in direction of the angular velocity ω_2.

The angular velocity vector ω2 changes due to the rotation of the cab, ω_1.

GIST

Sample Problem 2

Consider the following disk and rod.



Rod AB is 7 in long.

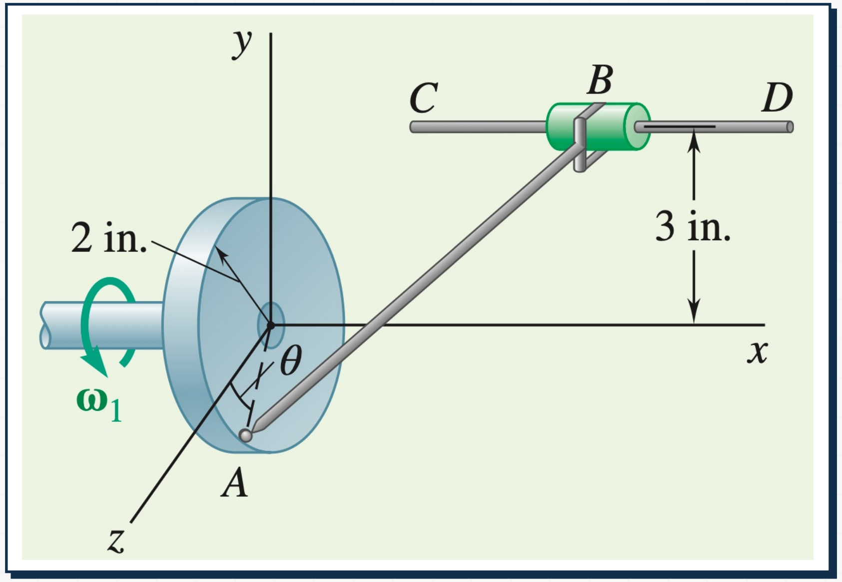
Rod AB is attached to the disk by a ball-socket joint and to the collar by a clevis.

A disk rotates in yz plane at a constant rate $\omega_1 = 12$ rad/s.

The collar is free to slide along CD horizontally.

Sample Problem 2

Consider the following disk and rod.



- Determine
  - The velocity of the collar @$\theta = 0°$.
  - The angular velocity of the rod @$\theta = 0°$.

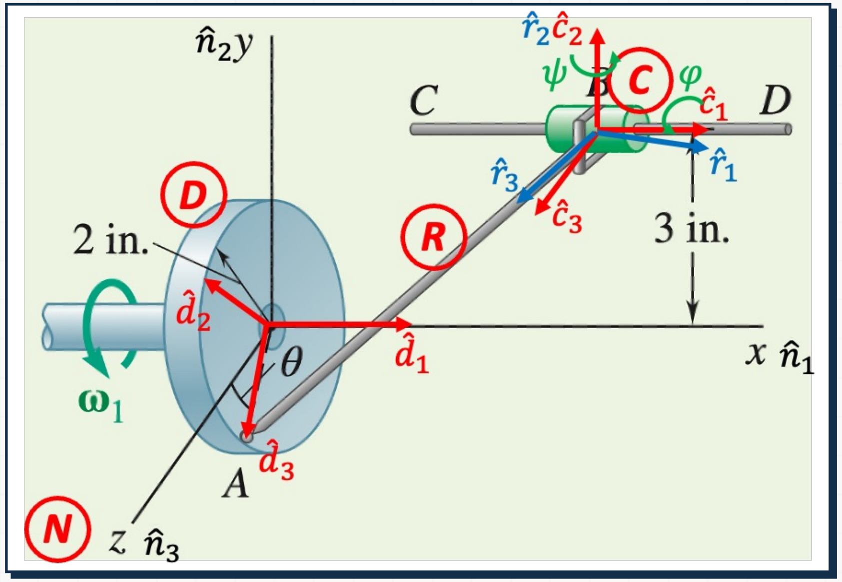
Sample Problem 2

Consider the following disk and rod.

- Strategy
  - Identify appropriate reference frames. It is useful to have only one degree of freedom between each reference frame to use the simple angular velocity.
  - Assign appropriate coordinate systems to each reference frame.
  - Build the direction cosine table.
  - Obtain the relative angular velocities between reference frames.
  - Use two-point relation to compute linear velocity of the collar and the angular velocity of the rod.

Sample Problem 2

Define reference frames and coordinate systems.



Inertial reference frame: N
with $\hat{n}_1\hat{n}_2\hat{n}_3$

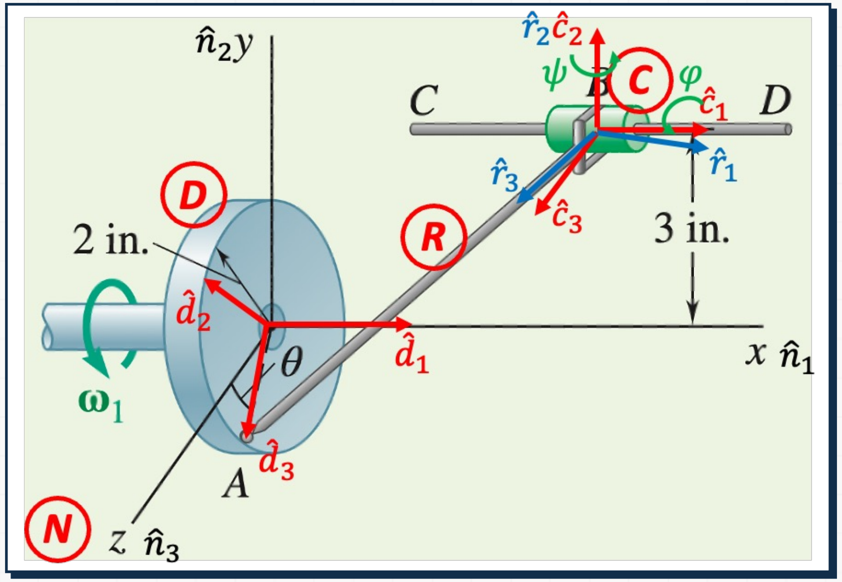
Reference frame of disk: D
with $\hat{d}_1\hat{d}_2\hat{d}_3$

Reference frame of rod: R
with $\hat{r}_1\hat{r}_2\hat{r}_3$

Reference frame of rod: C
with $\hat{c}_1\hat{c}_2\hat{c}_3$

Sample Problem 2

Build direction cosine tables.



	$\hat{d}_1$	$\hat{d}_2$	$\hat{d}_3$
$\hat{n}_1$	1	0	0
$\hat{n}_2$	0	$\cos\theta$	$-\sin\theta$
$\hat{n}_3$	0	$\sin\theta$	$\cos\theta$

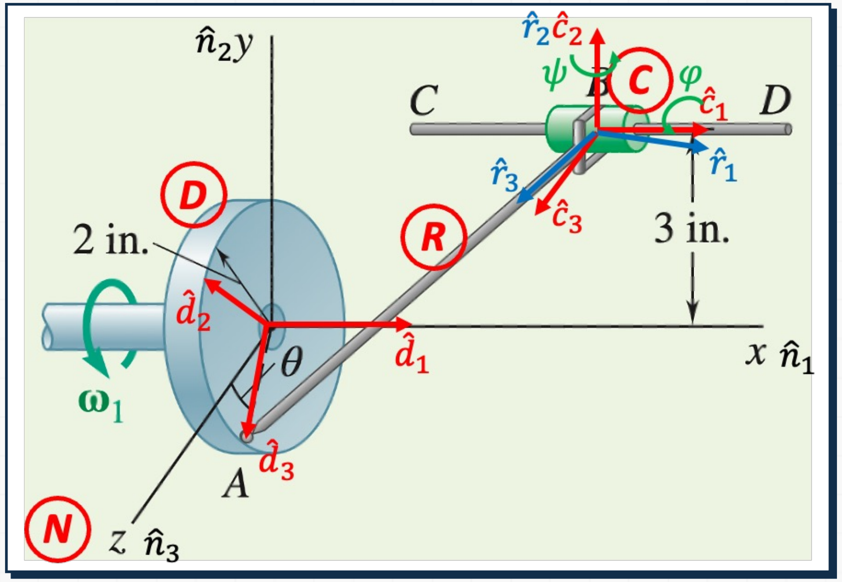
	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_3$
$\hat{c}_1$	$\cos\psi$	0	$\sin\psi$
$\hat{c}_2$	0	1	0
$\hat{c}_3$	$-\sin\psi$	0	$\cos\psi$

	$\hat{c}_1$	$\hat{c}_2$	$\hat{c}_3$
$\hat{n}_1$	1	0	0
$\hat{n}_2$	0	$\cos\varphi$	$-\sin\varphi$
$\hat{n}_3$	0	$\sin\varphi$	$\cos\varphi$

GIST

Sample Problem 2

Obtain relative angular velocities.



Disk rotates with angle θ about x axis with respect to the ground.

$$ {}^{N}\vec{\omega}^{D} = \omega_1 \hat{n}_1 = \omega_1 \hat{d}_1 $$

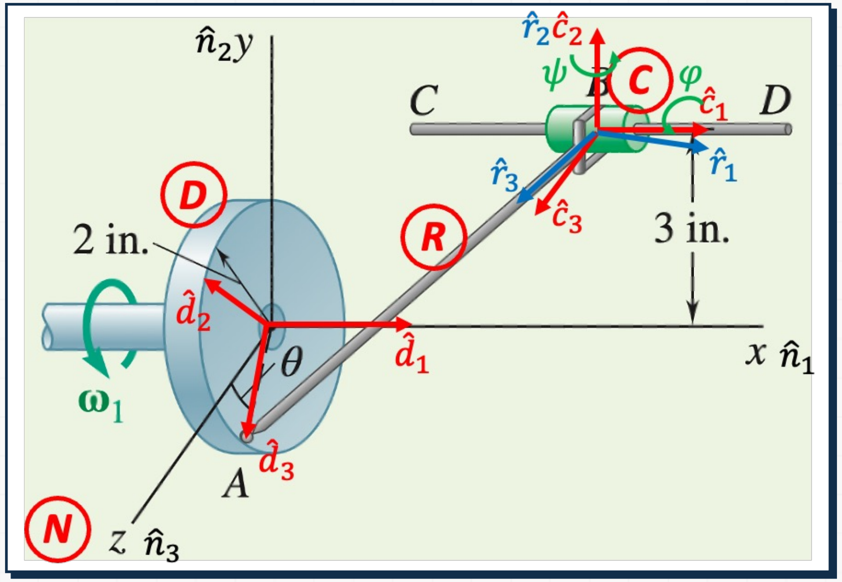
Collar rotates with angle φ about x axis with respect to the ground.

$$ {}^{N}\vec{\omega}^{C} = \dot{\varphi} \hat{c}_1 $$

GIST

Sample Problem 2

Obtain relative angular velocities.



Rod rotates with angle ψ about y axis with respect to the collar.

$$^{C}\vec{\omega}^{R} = \dot{\psi}\hat{c}_2$$

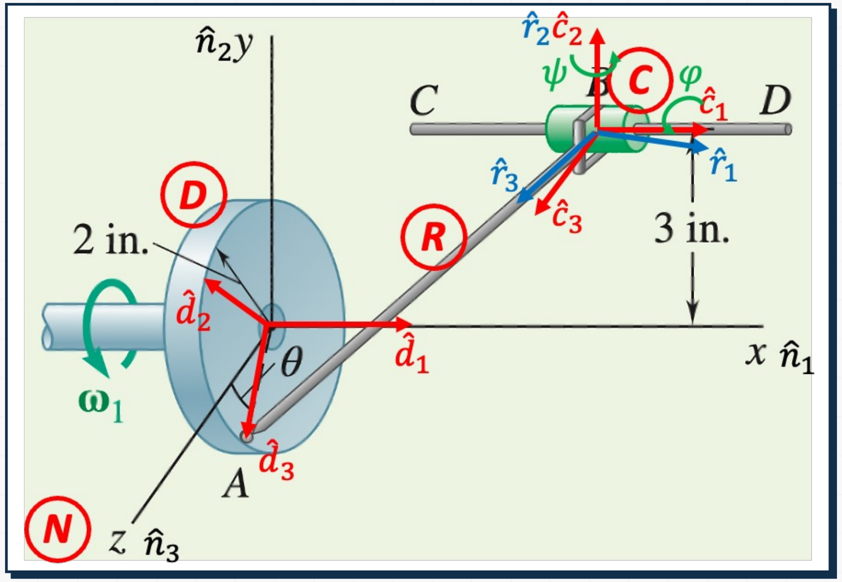
Angular velocity of rod with respect to the ground.

$$^{N}\vec{\omega}^{R} = {}^{N}\vec{\omega}^{C} + {}^{C}\vec{\omega}^{R} = \dot{\varphi}\hat{c}_1 + \dot{\psi}\hat{c}_2$$

GIST

Sample Problem 2

Compute the velocity of A and B.



Velocity of A from disk

$$^{N}\vec{v}^{A} = {}^{N}\vec{\omega}^{D} \times 2\hat{d}_3 = \omega_1\hat{d}_1 \times 2\hat{d}_3 = -2\omega_1\hat{d}_2$$

Velocity of B using two-point relation

$$\begin{aligned} ^{N}\vec{v}^{B} &= {}^{N}\vec{v}^{A} + {}^{N}\vec{\omega}^{R} \times \overrightarrow{AB} \\ &= -2\omega_1\hat{d}_2 + \left(\dot{\varphi}\hat{c}_1 + \dot{\psi}\hat{c}_2\right) \times (-7\hat{r}_3) \\ &= -7\cos\psi\,\dot{\psi}\hat{c}_1 \\ &= (-24\cos\varphi + 7\cos\psi\,\dot{\varphi})\hat{c}_2 \\ &= \left(24\sin\varphi + 7\sin\psi\,\dot{\psi}\right)\hat{c}_3 \end{aligned}$$

Both $\hat{c}_2$ and $\hat{c}_3$ components are zeros.

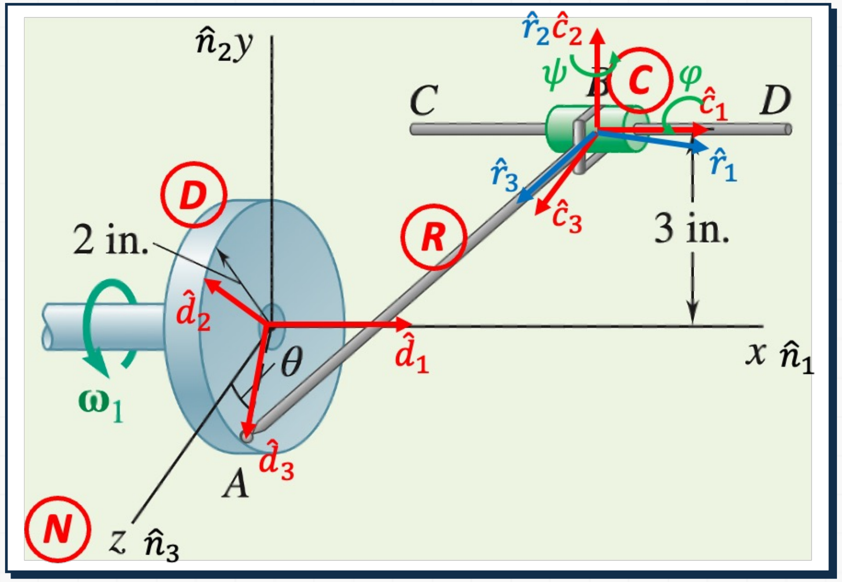
$$-24\cos\varphi + 7\cos\psi\,\dot{\varphi} = 0$$

$$24\sin\varphi + 7\sin\psi\,\dot{\psi} = 0$$

GIST

Sample Problem 2

Compute φ and ψ.



Assume that point C is at the y axis.
Then, consider a triangle ACB. Using law of cosine,

$$3^2 + 2^2$$
$$= 6^2 + 7^2 - 2 \cdot 6 \cdot 7 \cos(90° + \psi)$$
$$\psi = -1.03 rad$$

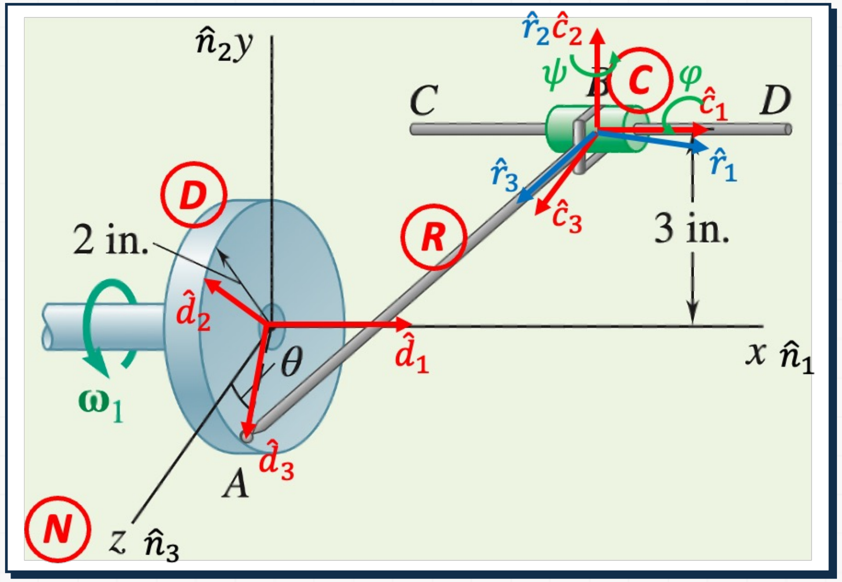
Consider the same triangle projected on yz plane.

$$\tan(90° - \varphi) = \frac{2}{3}$$
$$\varphi = 0.983 rad$$

$$\therefore \dot{\varphi} = 3.693 rad/s, \dot{\psi} = 3.328 rad/s$$

Sample Problem 2

Compute ${}^{N}\vec{\omega}^{R}$ and ${}^{N}\vec{v}^{B}$.



From,

$$-24\cos\varphi + 7\cos\psi\,\dot{\varphi} = 0$$
$$24\sin\varphi + 7\sin\psi\,\dot{\psi} = 0$$
$$\therefore\ \dot{\varphi} = 3.693 rad/s,\ \dot{\psi} = 3.328 rad/s$$

Linear velocity of collar

$$-7\cos\psi\,\dot{\psi}\hat{c}_1 = -11.993 in/s$$

Angular velocity of the rod

$${}^{N}\vec{\omega}^{R} = 3.693\hat{c}_1 + 3.328\hat{c}_2 rad/s$$

Sample Problem 2

Reflect and Think

Sometimes, obtaining simple angular velocity may not be straightforward.

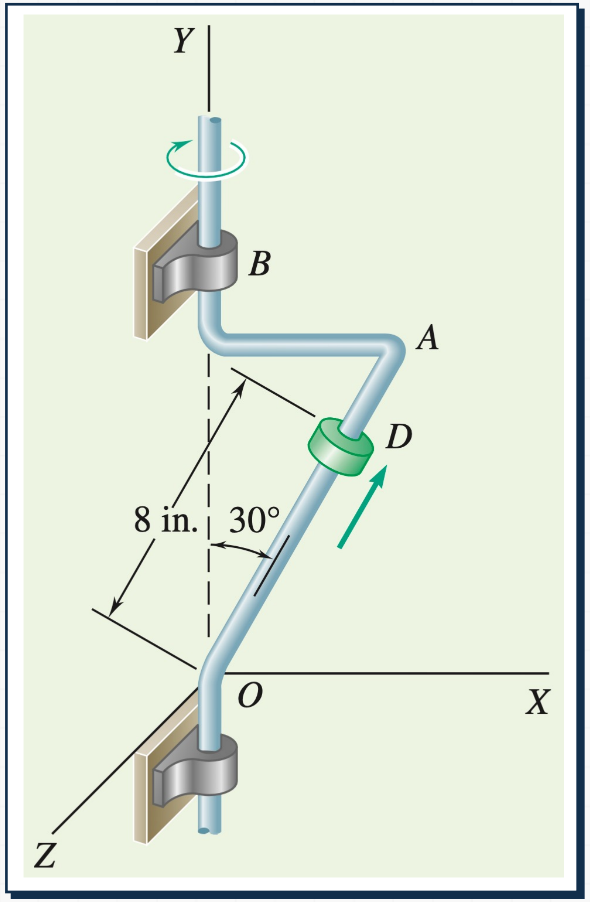
However, if constraints can be imposed, the problem can easily be solved.

Trigonometric relationship is used in many applications.

GIST

Sample Problem 3

Consider the following bent rod and collar.



The bent rod OAB rotates.

At the instant shown, $\omega = 20 rad/s$,
$\alpha = 200 rad/s^2$ in clockwise direction.

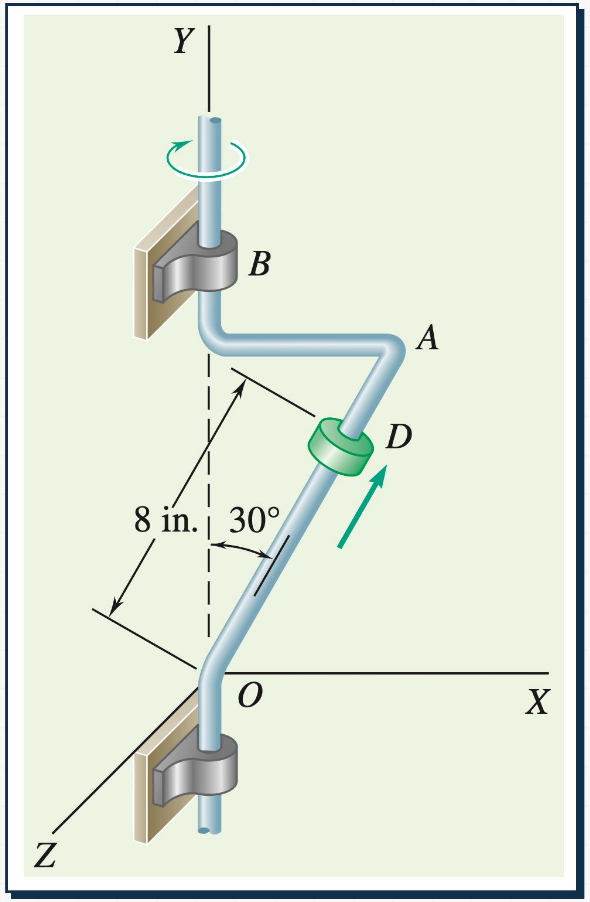
The collar D moves along the rod

At the instant shown, $v^D = 50 in/s$,
$a^D = 600 in/s^2$ upward.

GIST

Sample Problem 3

Consider the following bent rod and collar.



- Determine
  - The velocity of the collar.
  - The acceleration of the collar.

Sample Problem 3

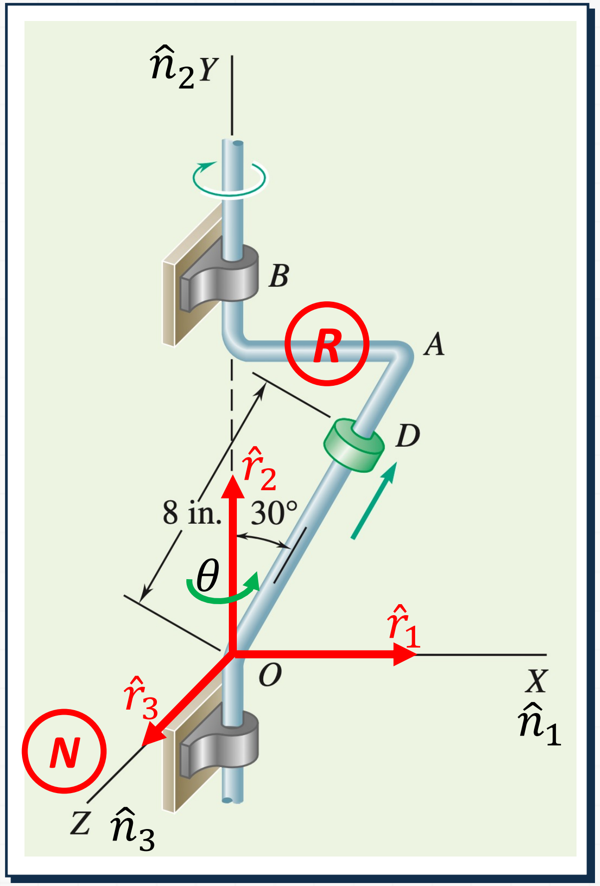
Consider the following bent rod and collar.

- Strategy
  - Identify appropriate reference frames. It is useful to have only one degree of freedom between each reference frame to use the simple angular velocity.
  - Assign appropriate coordinate systems to each reference frame.
  - Build the direction cosine table.
  - Obtain the relative angular velocities between reference frames.
  - Use one-point relation to compute linear velocity of the collar and the angular velocity of the collar.

GIST

Sample Problem 3

Define reference frames and coordinate systems.

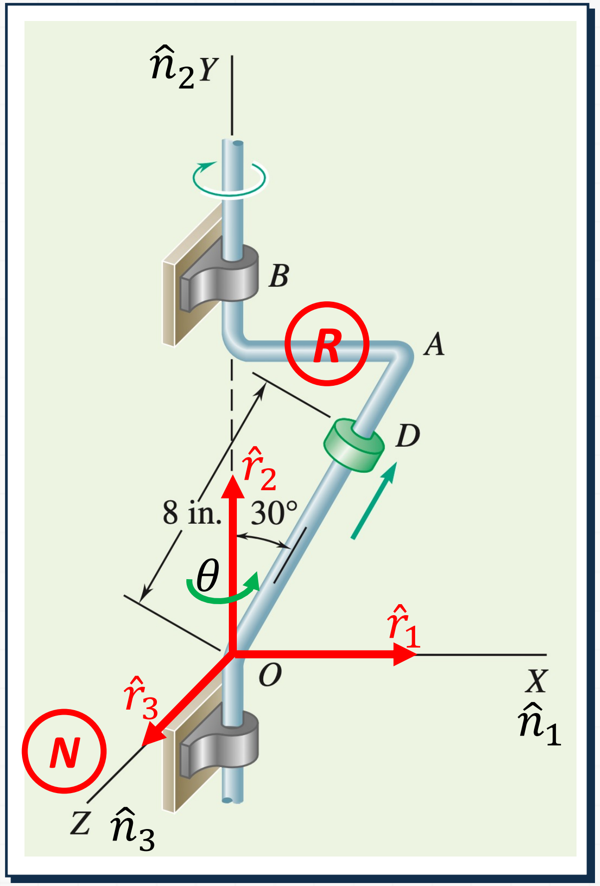


Inertial reference frame: N
with $\hat{n}_1\hat{n}_2\hat{n}_3$

Reference frame of rod: R
with $\hat{r}_1\hat{r}_2\hat{r}_3$

Sample Problem 3

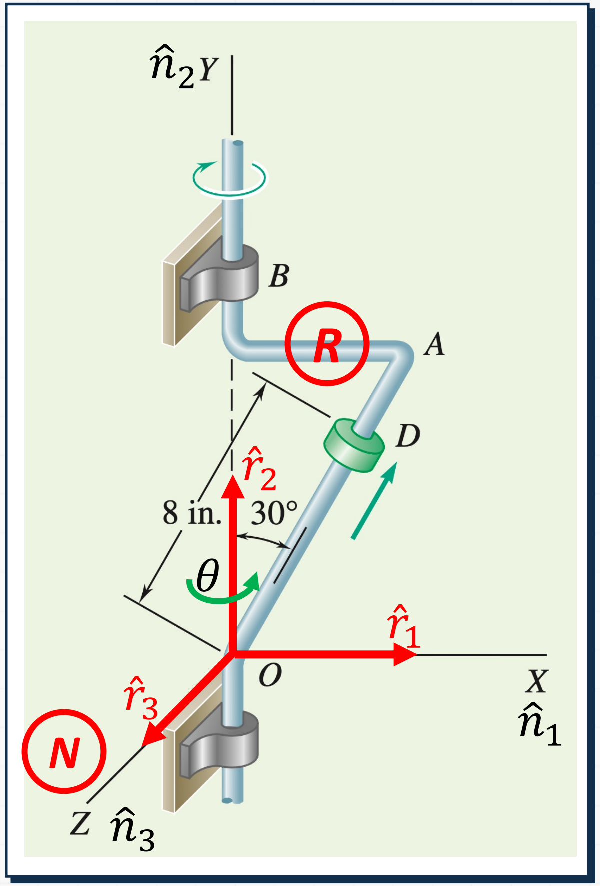
Build direction cosine tables.



	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_3$
$\hat{n}_1$	$\cos\theta$	0	$\sin\theta$
$\hat{n}_2$	0	1	0
$\hat{n}_3$	$-\sin\theta$	0	$\cos\theta$

Sample Problem 3

Obtain relative angular velocity and angular acceleration.



Rod rotates with angle θ about y axis with respect to the ground.

$$^{N}\vec{\omega}^{R} = \dot{\theta}\hat{n}_2 = \dot{\theta}\hat{r}_2$$

$$^{N}\vec{\alpha}^{R} = \frac{^{N}d}{dt}\left(^{N}\vec{\omega}^{R}\right)$$

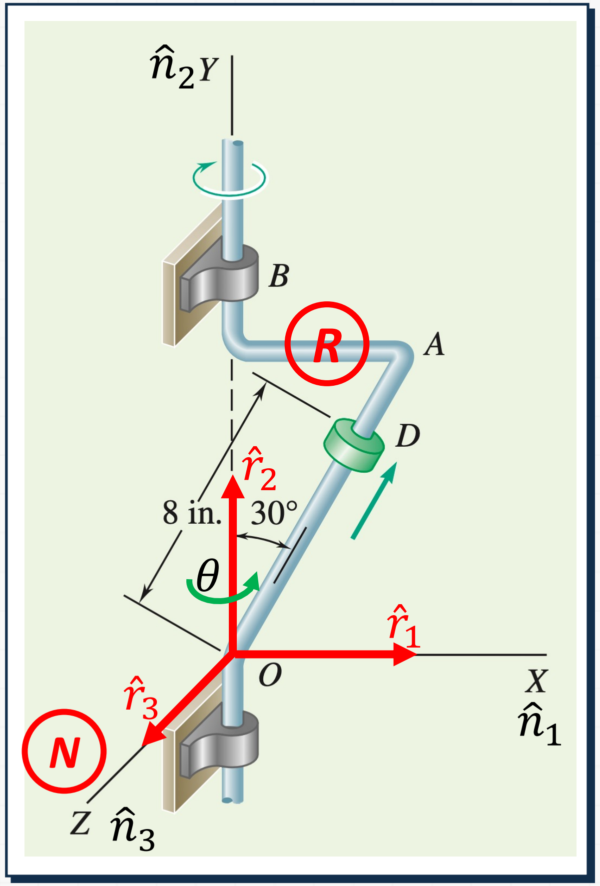
$$= \frac{^{R}d}{dt}\left(^{N}\vec{\omega}^{R}\right) + {^{N}\vec{\omega}^{R}} \times {^{N}\vec{\omega}^{R}}$$

$$= \frac{^{R}d}{dt}\left(^{N}\vec{\omega}^{R}\right)$$

$$= \ddot{\theta}\hat{r}_2$$

Sample Problem 3

Compute the velocity of D.



Compute the velocity of point D^* using two-point relation.

$$
\begin{aligned}
{}^{N}\vec{v}^{D^*} &= {}^{N}\vec{v}^{O} + {}^{N}\vec{\omega}^{R} \times \overrightarrow{OD^*} \\
&= 0 + \dot{\theta}\hat{r}_2 \times \\
&\quad 8(\sin 30^\circ\,\hat{r}_1 + \cos 30^\circ\,\hat{r}_2) \\
&= -8\sin 30^\circ\,\dot{\theta}\hat{r}_3
\end{aligned}
$$

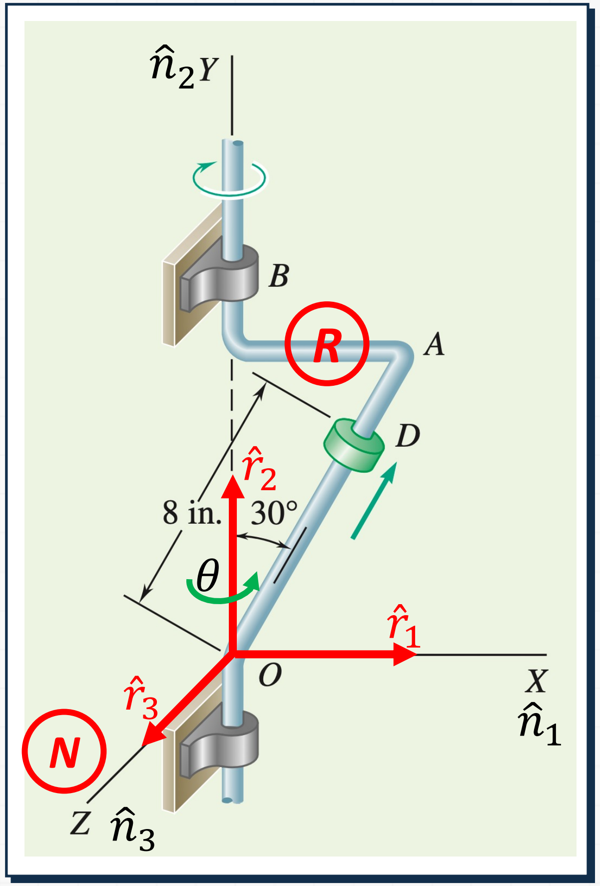
Compute the velocity of point D using one-point relation.

$$
\begin{aligned}
{}^{N}\vec{v}^{D} &= {}^{N}\vec{v}^{D^*} + {}^{R}\vec{v}^{D} \\
&= -8\sin 30^\circ\,\dot{\theta}\hat{r}_3 \\
&\quad +50(\sin 30^\circ\,\hat{r}_1 + \cos 30^\circ\,\hat{r}_2) \\
&= 25\hat{r}_1 + 43.3\hat{r}_2 + 80\hat{r}_3
\end{aligned}
$$

GIST

Sample Problem 3

Compute the acceleration of D.

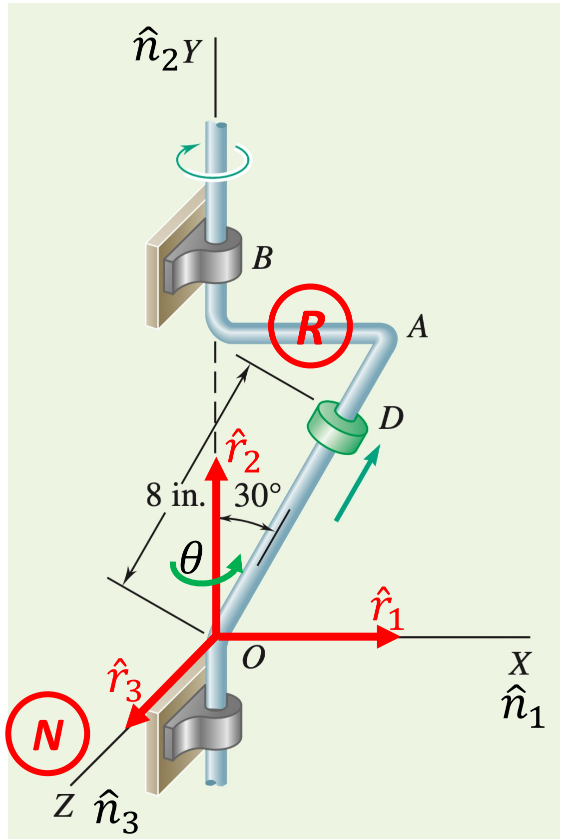


Compute the acceleration of point D^* using two-point relation.

$$
\begin{aligned}
{}^N\vec{a}^{D^*} &= {}^N\vec{a}^{O} + {}^N\vec{\alpha}^{R} \times \overrightarrow{OD^*} \\
&\quad + {}^N\vec{\omega}^{D} \times \left({}^N\vec{\omega}^{R} \times \overrightarrow{OD^*} \right) \\
&= 0 + \ddot{\theta}\hat{r}_2 \times 8(\sin 30°\,\hat{r}_1 \\
&\quad + \dot{\theta}\hat{r}_2 \times \dot{\theta}\hat{r}_2 \times 8(\sin 30°\,\hat{r}_1 \\
&= -1600\hat{r}_1 + 800\hat{r}_3 in/s^2
\end{aligned}
$$

Sample Problem 3

Compute the acceleration of D.



Compute the acceleration of point D using one-point relation.

$$
\begin{aligned}
{}^{N}\vec{a}^{D} &= {}^{N}\vec{a}^{D^*} + {}^{R}\vec{a}^{D} + 2\,{}^{N}\vec{\omega}^{R} \times {}^{R}\vec{v}^{D} \\
&= -1600\hat{r}_1 + 800\hat{r}_3 \\
&\quad + 600(\sin 30^\circ\,\hat{r}_1 + \cos 30^\circ\,\hat{r}_2) \\
&\quad -40\hat{r}_2 \times 50(\sin 30^\circ\,\hat{r}_1 \\
&= -1300\hat{r}_1 + 519.6\hat{r}_2 + 1800\hat{r}_3 in
\end{aligned}
$$

GIST

Sample Problem 3

Reflect and Think

Reference frame can be assigned to a moving collar. When the collar is moving on a reference frame R, it experiences a Coriolis acceleration due to angular velocity in 3D space.

GIST

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, McGraw Hill, Chapter 15
- 유홍희, "동역학", 1판, 문운당, 8장